Building an AI-Powered Spatial Query Widget with Experience Builder and FastAPI

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Background

AI is the Current It Girl

- Push for AI
- Staff continues to want simpler and faster tools
- Training the public to use GIS tools is difficult
- “Simple” GIS queries in natural language really do sound simple

Initial Ideas

Easiest Use Case

- Spatial Analysis using 311 or PD data
- Geocoding Error Detection
- Map or Report Creation
- Traffic Incident Hotspots
- AI Imagery Analysis and Permitting

Spatial Analysis AI Prompt Goals

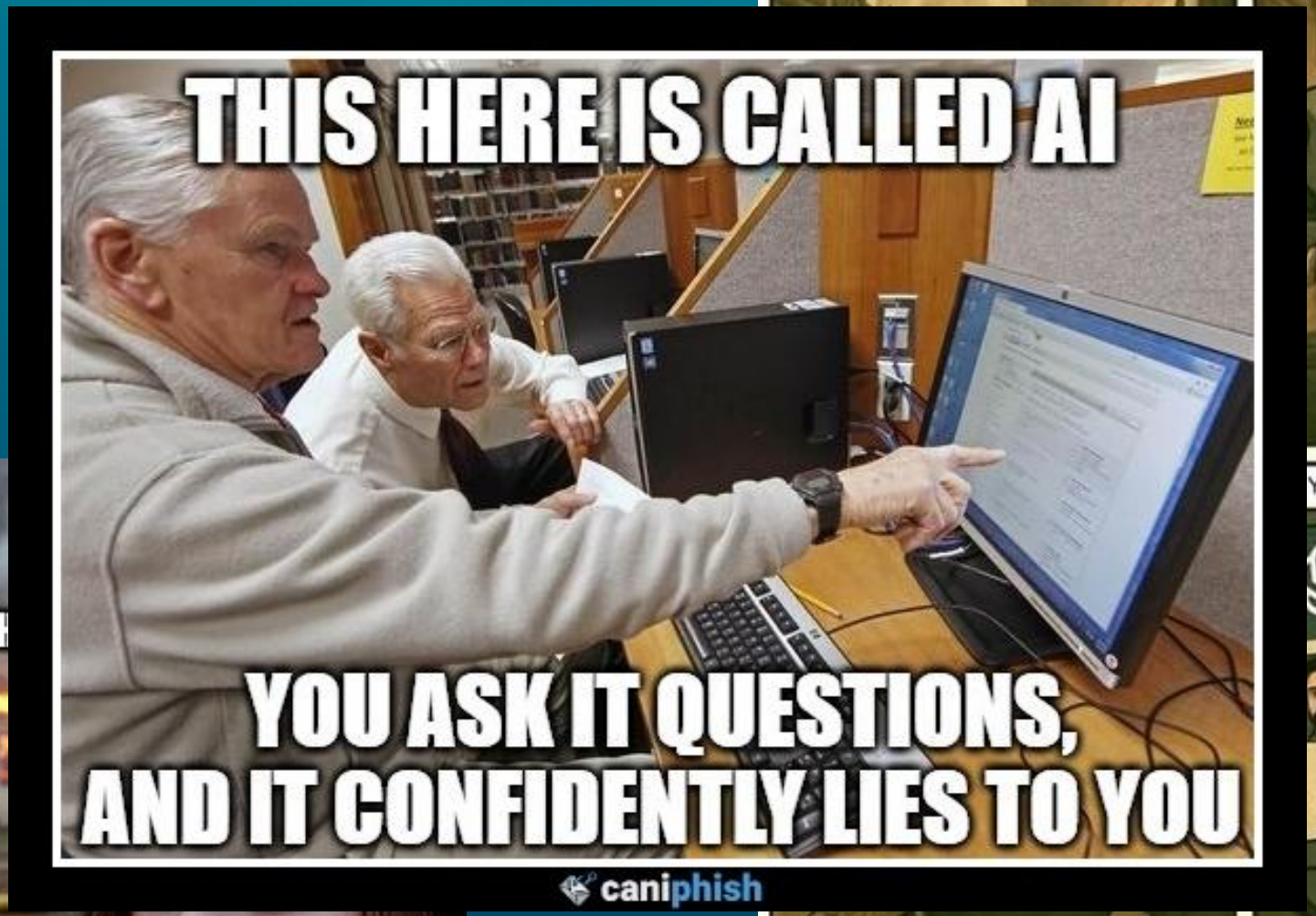
We Do This Not Because It's Easy, But Because We Thought It Would Be

- Create a single widget where users can simply ask:
  - “Show cases from the last week within 600 feet of all schools”
- The system would then:
  - Interpret the request with AI
  - Run the spatial analysis in Python
  - Return the features
  - Filter and select them in the map and table

Step 1

Duh

- Ask the BFF



Widget Workflow

K.I.S.S for the User, Not for Me

- User enters natural language query
- Widget sends request to FastAPI API
- Python interprets query with AI
- Spatial analysis runs
- Results returned to widget
- Widget Filters map layer and table

Architecture

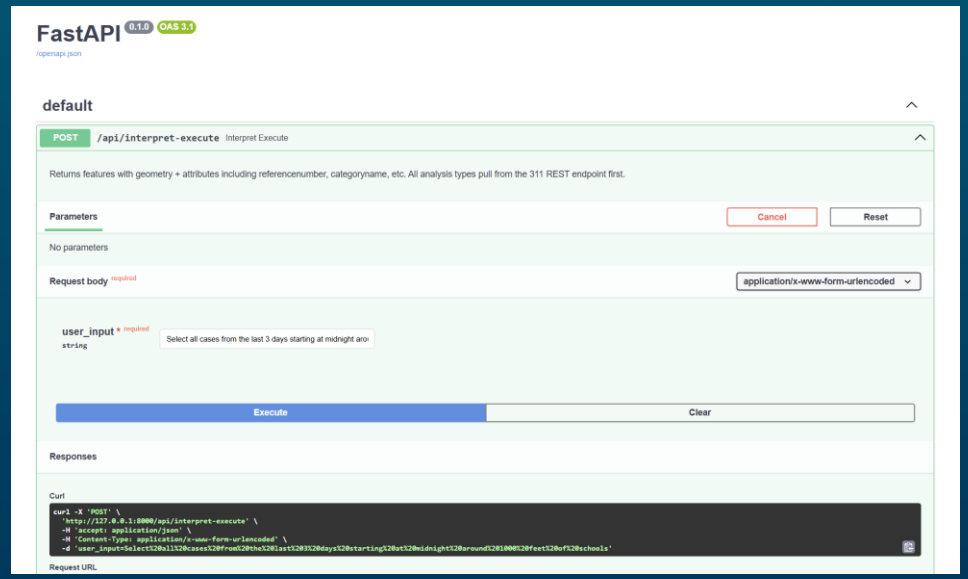
What Does What…

- Frontend:
  - ArcGIS Experience Builder custom widget
  - React and Typescript
- Backend:
  - FastAPI Service
  - OpenAI or ClaudeAPI for NLP
  - GeoPandas for spatial analysis

FastAPI Backend

This is Pretty Cool

- FastAPI handles the API endpoint:
  - POST /api/interpret-execute



- Responsibilities:
  - Receive user query
  - Ask AI to interpret parameters
  - Query ArcGIS feature services
  - Run spatial analysis
  - Return features to the widget

Using AI for Query Interpretation

Tell Me What They Want

- OpenAI/Claude converts NLP into structured parameters

```
prompt = f"""
Current local datetime is {current_local_dt_str} in the user's local timezone (America/Los_Angeles).
Analyze this user request: {user_input}

Return ONLY valid JSON with:
- "analysis_type": one of "clustering", "proximity", or "hotspot"
- "buffer_distance": number (feet) or null
- "start_datetime": "YYYY-MM-DD HH:MM:SS AM/PM" or null
- "end_datetime": "YYYY-MM-DD HH:MM:SS AM/PM" or null
- "school_type": one of "Private", "Public", or "Charter" or null

Interpret relative time phrases like:
- "today"
- "yesterday"
- "last 24 hours"
- "last 36 hours"
- "last week"

relative to this local datetime: {current_local_dt_str}.

Example:
{{"analysis_type":"proximity","buffer_distance":500,"start_datetime":"2026-03-05 08:15:00","end_datetime":"2026-03-06 20:15:00","school_type":"Public"}}
"""
```

Python Spatial Analysis

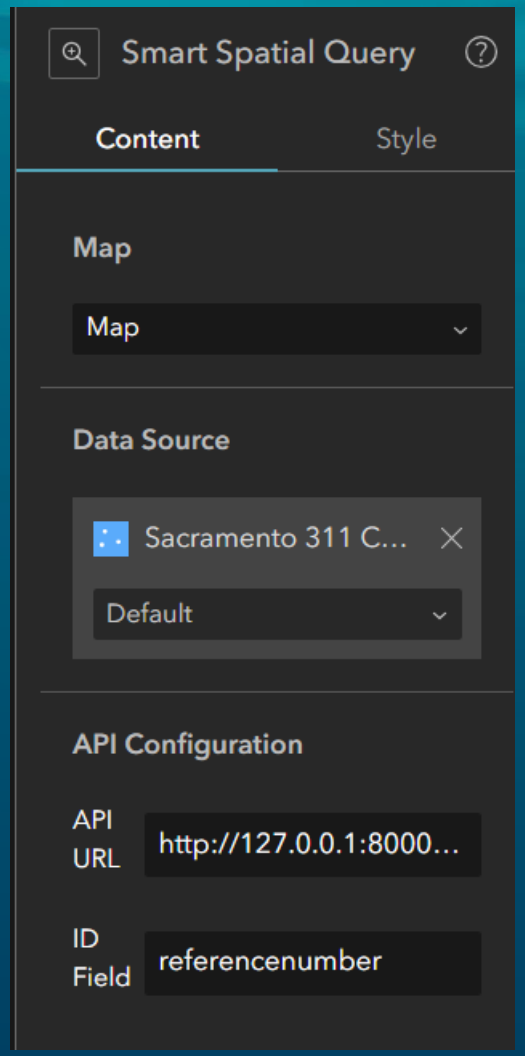
The Harder Part Surprisingly

- Python handles the spatial processing:
  - Query Feature Services
  - Convert to GeoDataFrame
  - Buffer school locations
  - Spatial join with 311 incidents
  - Return matching features

Filtering ExB Map and Table

Also, Difficult

- Custom Experience Builder Widget:
  - Builds WHERE clause from returned IDs
  - Applies map filter to FeatureLayerView
  - Updates data source query
  - Selects matching records
- Result: Map + Table update automatically



Challenges and Pain Points

<Insert Bad Words>

- Main issues encountered:
  - AI tries but it really needs checking and hand holding
  - Projection mismatches (3857 vs 2226)
  - GeoPandas geometry column differences
  - Feature service query performance
  - Parsing time ranges from natural language

Benefits of the NLP Interface

AI's Not So Bad…

- NLP queries make spatial analysis easier
  - Reduces training required
  - Eliminates multi-widget workflows
  - More intuitive for non-GIS users
  - Faster data discovery

Future Improvements

All We Need is Time

- Include additional parameters for AI to find
- Expand on the spatial analysis types
- Expand to other datasets and AI models
- Create actual functions to run the API calls rather than python

Demo

ArcGIS Experience Builder | Untitled experience 1 | FastAPI - Swagger UI | Gemini
127.0.0.1:8000/docs#/default/interpret_execute_api_interpret_execute_post
SPD Gerrit | Incognito City | Incognito PD | PSIT Applications D... | PD Enterprise | GIS | Server Tools | Information Technol... | Intranet | Microsoft Office 365 | Power BI | Home - SumTotal | Personal | All Bookmarks

FastAPI 0.1.0 OAS 3.1

/openapi.json

default

POST `/api/interpret-execute` Interpret Execute

Returns features with geometry + attributes including referencenumber, categoryname, etc. All analysis types pull from the 311 REST endpoint first.

Parameters Try it out

No parameters

Request body required application/x-www-form-urlencoded

user_input * required
string

Responses

Code	Description	Links
200	Successful Response Media type: application/json Controls Accept header. Example Value | Schema `"string"`	*No links*

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ArcGIS Experience Builder | Untitled experience 1 | FastAPI - Swagger UI

Not secure https://localhost:3001/experience/1?draft=true

SPD Gerrit | Incognito City | Incognito PD | PSIT Applications D... | PD Enterprise | GIS | Server Tools | Information Technol... | Intranet | Microsoft Office 365 | Power BI | Home - SumTotal | Personal | All Bookmarks

Double click to edit text

Subtitle

Find address or place

Sacramento 311 Calls View (Current)

Date Created is in the last

Amount | Day

Schools

SCHOOL_TYPE is

Public

Cleared.

API: 0 | Table: 0 | Selected: 0

Select all cases from the last 3 days starting at midnight around 700 feet of all publ...

Run | Clear

Esri, NASA, NGA, USGS, FEMA | County of Sacramento, California State Parks, Esri, TomTom, Garmin, SafeGraph, GeoTechnologies, Inc, METI/NASA, USGS, Bureau of Land Managem... USDA, USFWS | GIS IT

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